Data Cleaning and Preprocessing

Contents

Experiment 1: Identifying missing values in sensor and API logs

Goal

To detect and quantify missing, null entries in previously acquired datasets (eg. from Thinkspeak API, OpenEI pressure logs, etc)

Rationale

Sensor data or API feeds often contains partial, intermittent signals - which leads to missing values with respect to timestamps or certain fields.

Topics covered

MCAR (Missing Completely at Random), MAR (Missing at Random), MNAR (Missing Not At Random), isnull(), info(), .sum(), dropna (row/column deletion)

Methodology

1. Load previously acquired dataset from Experiment 1 (eg. geothermal pressure, IoT strain, etc)
2. Detect missing entries using .isnull(), .notna(), .info()
3. Identify missing timestamps or zero readings as “structurally missing”

Expected Results

1. List of affected records
2. Summary of affected columns
3. Recognition of sensor-specific failure patterns.

Experiment 2: Imputing Missing Weather data from Web Scraping

Goal

Apply imputation on structured and semi-structured weather data scraped in earlier labs.

Rationale

Scraped HTML often leads to missing or incorrect entries which occurs due to broken tags, failed fetches, etc. Such anomalies simulate real-world challenges while developing engineering dashboards or other analysis.

Topics covered

dropna, imputation - mean, median, etc

Methodology

1. Use the weather data from Experiment 1
2. Introduce artificial missing values and identify failed parses.
3. Impute using .fillna() with default placeholders, mode or forward fill (ffill).

Expected Results

1. Cleaned data with no NaNs
2. Differences between HTML-based and API-based reliability
3. Summary of how many entries were recovered.

Experiment 3: Time-series Interpolation in IoT sensor Streams

Goal

To restore continuous sensor readings in time-series datasets using interpolation.

Rationale

In real-time systems, dropped packets or delayed sensors reads results in NaN timestamps or missing values.

Topics covered

KNN interpolation, realworld sensor gaps and noise

Methodology

1. Load time-series from “Strain Sensor or Hydraulic Pressure Logs” 2. Introduce missing entries at random time intervals (simulate packet loss). 3. Apply .interpolate (method = “time”) or spline, and plot before-after.

Expected Results

1. Restored continuous time-series
2. Plots of original vs. imputed signals
3. Comparison of different interpolation methods.

Experiment 4: KNN based imputation in a real-time temperature feed

Goal

To implement KNN based imputation method on a real-time multi-variate dataset (eg. temperature, humidity from an API feed)

Rationale

Statistical and Temporal methods may not work when multiple correlated fields are involved, KNN based imputation method leverages multivariate similarity.

Topics covered

Imputation methods - mean, median, etc, KNN Interpolation, Real world sensor gaps and noise, Contextual imputation

Methodology

1. Load the dataset acquired from wttr.in APi with temp, humidity and wind.
2. Introduce 10-15% missing values randomly
3. Apply KNNImputer from sklearn.impute.
4. Plot the values before and after imputation.

Expected Results

1. Clean dataset restored using KNN.
2. Visualization of missing vs. imputed areas.
3. Observation of correlation-based filling.

Experiment 5: Visual Diagnostics of Missingness Patterns

Goal

Implement visualization techniques such as heatmap and matrix based visualizations to explore the missing values structure across the data.

Rationale

Visualing is a powerful diagnostic tool in early stages of developing engineering data pipelines.

Topics covered

MCAR (Missing Completely at Random), MAR (Missing at Random), MNAR (Missing Not At Random), seaborn library, missing no, contextual imputation.

Methodology

1. Use missingno.matrix() and seaborn.heatmap() on any two datasets with structured missing data.
2. Generate comparative plots.
3. Annotate spatial patterns (eg. block missingness, time drift).

Expected Results

1. Indicate Visual patterns involving systemic issues (eg. missing in bursts, sensor blackouts)

Experiment 6: Comparing Model accuracy on Raw vs. Imputed Sensor Data

Goal

To evaluate the impact of missing value handling on predictive performance

Rationale

In embedded systems or automation related applications, bad handling of missing data can lead to degradation in the quality of prediction, control or alerting systems.

Topics Covered

Impact of missing data on modeling

Methodology

1. Use a dataset (eg. vibration signals or strain reading with classification labels: Normal vs. faulty).
2. Train a simple classifier (eg. decision tree) on:
    a. Raw (Nan) data
    b. Mean-Imputed data
    c. KNN-imputed data
3. Compare accuracy, confusion matrix.

Expected Results

1. Visualize how treating missing data affects the model's performance

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